

ShopOpt

Online Shopping Product Optimizer

Project Report · Smart basket selection within a budget

1. Overview

ShopOpt is a web application that helps a shopper build the smartest basket within a fixed budget. It presents a catalogue of branded products across six sections, lets the shopper pick an objective (rating, utility, discount or a weighted combination) and then solves the selection with the **0/1 Knapsack** algorithm using dynamic programming. The app also includes a Sorting Lab (bubble, selection, insertion and merge sort), a Binary Search demo on prices, an analytics page that measures algorithm run-times, special offers, voice search in 20 languages, and optional login with a backend API (with a demo mode when the server is unavailable).

32	6	₹10,000	20
Products in catalogue	Sections	Default budget	Voice-search languages

Application pages

Page	Purpose
Dashboard	Budget, product count, best value found and total savings at a glance.
Products	Browse and search by section, select items, add or remove products.
Special offers	Section-wise promo codes and a list of hot deals (25% off or more).
Optimizer	Choose an objective and budget, then run the live 0/1 Knapsack visualisation.
Sorting Lab	Animate bubble, selection, insertion and merge sort with comparison and swap counts.
Binary Search	Search a price in the merge-sorted price list, step by step.
Analytics	In-browser timing of knapsack, sorting and binary search.
Complexity	Time and space complexity summary of the algorithms used.

2. Product catalogue

Each product carries a price (a multiple of ₹100), a rating out of 5, a utility score (0–100) and a discount percentage. The first twelve products are listed under Electronics, except the Laptop Backpack, which the app files under Clothing.

#	Product	Brand	Section	Price	Rating	Utility	Discount
1	Watch SE	Apple	Electronics	₹4,000	4.4	95	10%
2	WH-CH520 Headphones	Sony	Electronics	₹2,500	4.6	90	20%
3	K2 Mechanical Keyboard	Keychron	Electronics	₹3,000	4.7	85	15%
4	G102 Gaming Mouse	Logitech	Electronics	₹1,500	4.5	75	25%
5	Flip 6 Speaker	JBL	Electronics	₹2,000	4.3	70	30%
6	C270 HD Webcam	Logitech	Electronics	₹1,500	4.1	60	12%
7	20000mAh Power Bank	Xiaomi	Electronics	₹1,000	4.5	55	18%
8	6-in-1 USB-C Hub	UGREEN	Electronics	₹1,000	4.2	50	22%
9	Aluminium Laptop Stand	Portronics	Electronics	₹1,500	4.6	65	8%
10	LED Desk Lamp	Philips	Electronics	₹1,000	4.0	40	35%
11	Smart Band 8	Xiaomi	Electronics	₹2,500	4.2	72	28%
12	Laptop Backpack	Wildcraft	Clothing	₹2,000	4.5	68	16%
13	Air Max Sneakers	Nike	Clothing	₹4,000	4.5	88	30%
14	511 Slim Jeans	Levi's	Clothing	₹2,500	4.3	70	25%
15	Cotton T-Shirt	H&M;	Clothing	₹800	4.2	55	40%
16	Puffer Jacket	Zara	Clothing	₹3,500	4.4	82	35%
17	Matte Lipstick	Lakmé	Beauty	₹600	4.4	60	30%
18	Hydrating Moisturiser	Nivea	Beauty	₹500	4.5	58	20%
19	Eau de Parfum 100ml	Park Avenue	Beauty	₹2,500	4.6	75	25%
20	Kajal Pencil	Maybelline	Beauty	₹300	4.2	50	35%
21	Whole Wheat Atta 5kg	Aashirvaad	Groceries	₹300	4.6	80	10%
22	Full Cream Milk 1L	Amul	Groceries	₹100	4.7	70	5%
23	Sunflower Oil 1L	Fortune	Groceries	₹200	4.3	62	12%
24	Basmati Rice 5kg	India Gate	Groceries	₹600	4.5	78	18%
25	Paracetamol 650 Strip	Dolo	Medicines	₹100	4.5	85	10%
26	Antiseptic Liquid 500ml	Dettol	Medicines	₹300	4.6	70	15%
27	Pain Relief Spray	Volini	Medicines	₹400	4.4	65	20%
28	Digital Thermometer	Omron	Medicines	₹500	4.5	60	18%
29	Dry Dog Food 3kg	Pedigree	Pets	₹1,500	4.6	85	15%
30	Cat Food Pouches	Whiskas	Pets	₹500	4.5	70	10%
31	Chew Toy Bone	Kong	Pets	₹300	4.2	50	25%
32	Grooming Brush	Trixie	Pets	₹400	4.3	45	20%

3. Special offers

Section	Offer	Details	Code
Clothing	Flat 30% off sneakers & jackets	Seasonal sale	STYLE30
Electronics	Up to 25% off audio & wearables	Headphones, speakers, watches	TECH25
Beauty	Buy 2, get 15% off	On all beauty products	GLOW15
Groceries	₹100 off on ₹1,000+	Staples, dairy and oils	FRESH100
Medicines	Free delivery on health essentials	Orders above ₹300	CARE300
Pets	20% off first pet food order	Dog and cat food	PAWS20

Logged-in members get an extra 5% off with code **MEMBER5**. Offers are demo content for this prototype.

Hot deals (25% off or more) - 12 products

Product	Section	Price	Discount
G102 Gaming Mouse	Electronics	₹1,500	25%
Flip 6 Speaker	Electronics	₹2,000	30%
LED Desk Lamp	Electronics	₹1,000	35%
Smart Band 8	Electronics	₹2,500	28%
Air Max Sneakers	Clothing	₹4,000	30%
511 Slim Jeans	Clothing	₹2,500	25%
Cotton T-Shirt	Clothing	₹800	40%
Puffer Jacket	Clothing	₹3,500	35%
Matte Lipstick	Beauty	₹600	30%
Eau de Parfum 100ml	Beauty	₹2,500	25%
Kajal Pencil	Beauty	₹300	35%
Chew Toy Bone	Pets	₹300	25%

4. Optimizer - 0/1 Knapsack results

For a budget of ₹10,000 the knapsack table has one row per product and one column per ₹100 slot. Each cell keeps the better of include (product value + best value for the remaining budget) or exclude (value of the previous row). The optimal basket is then traced back from the last cell. Results below use the default weights for the combined objective (utility 40%, rating 30%, discount 30%).

Summary across objectives (budget ₹10,000)

Objective	Maximum value	Total cost	Remaining	Products	Savings
Utility	1173	₹9,900	₹100	18	₹2,082
Rating	1602	₹9,900	₹100	18	₹1,827
Discount	378	₹9,900	₹100	18	₹2,377
Combined value	1158	₹9,900	₹100	18	₹2,252

Each run compared 3,333 budget-and-product states (33 rows × 101 columns), covering every combination implicitly.

Selected basket - objective: Utility

Product	Section	Price	Value	Rating	Discount
G102 Gaming Mouse	Electronics	₹1,500	75	4.5	25%
20000mAh Power Bank	Electronics	₹1,000	55	4.5	18%
Cotton T-Shirt	Clothing	₹800	55	4.2	40%
Matte Lipstick	Beauty	₹600	60	4.4	30%
Hydrating Moisturiser	Beauty	₹500	58	4.5	20%
Kajal Pencil	Beauty	₹300	50	4.2	35%
Whole Wheat Atta 5kg	Groceries	₹300	80	4.6	10%
Full Cream Milk 1L	Groceries	₹100	70	4.7	5%
Sunflower Oil 1L	Groceries	₹200	62	4.3	12%
Basmati Rice 5kg	Groceries	₹600	78	4.5	18%
Paracetamol 650 Strip	Medicines	₹100	85	4.5	10%
Antiseptic Liquid 500ml	Medicines	₹300	70	4.6	15%
Pain Relief Spray	Medicines	₹400	65	4.4	20%
Digital Thermometer	Medicines	₹500	60	4.5	18%
Dry Dog Food 3kg	Pets	₹1,500	85	4.6	15%
Cat Food Pouches	Pets	₹500	70	4.5	10%
Chew Toy Bone	Pets	₹300	50	4.2	25%
Grooming Brush	Pets	₹400	45	4.3	20%
Total (18 products)		₹9,900	1173		

Selected basket - objective: Combined value

Product	Section	Price	Value	Rating	Discount
G102 Gaming Mouse	Electronics	₹1,500	72	4.5	25%
LED Desk Lamp	Electronics	₹1,000	61	4.0	35%
Cotton T-Shirt	Clothing	₹800	71	4.2	40%
Matte Lipstick	Beauty	₹600	68	4.4	30%
Hydrating Moisturiser	Beauty	₹500	62	4.5	20%
Kajal Pencil	Beauty	₹300	66	4.2	35%
Whole Wheat Atta 5kg	Groceries	₹300	66	4.6	10%
Full Cream Milk 1L	Groceries	₹100	59	4.7	5%
Sunflower Oil 1L	Groceries	₹200	58	4.3	12%
Basmati Rice 5kg	Groceries	₹600	69	4.5	18%
Paracetamol 650 Strip	Medicines	₹100	67	4.5	10%
Antiseptic Liquid 500ml	Medicines	₹300	65	4.6	15%
Pain Relief Spray	Medicines	₹400	64	4.4	20%
Digital Thermometer	Medicines	₹500	62	4.5	18%
Dry Dog Food 3kg	Pets	₹1,500	71	4.6	15%
Cat Food Pouches	Pets	₹500	61	4.5	10%
Chew Toy Bone	Pets	₹300	60	4.2	25%
Grooming Brush	Pets	₹400	56	4.3	20%
Total (18 products)		₹9,900	1158		

Why these products were selected

- The total cost stays within the budget.
- The basket maximises the chosen objective.
- Every selected product adds to the final value.
- Dynamic programming guarantees the result is optimal, unlike a greedy pick of the best-value items.

5. Sorting Lab

The four sorting algorithms were run on the 32 product prices (lowest → highest). Counts show how much work each one does; merge sort needs far fewer comparisons than the quadratic methods.

Algorithm	Comparisons	Swaps / writes	Time complexity	Space
Bubble sort	496	353	$O(n^2)$	$O(1)$
Selection sort	496	26	$O(n^2)$	$O(1)$
Insertion sort	376	353	$O(n^2)$	$O(1)$
Merge sort	117	160	$O(n \log n)$	$O(n)$

Sorted price list

₹100, ₹100, ₹200, ₹300, ₹300, ₹300, ₹300, ₹400, ₹400, ₹500, ₹500, ₹500, ₹600, ₹600, ₹800, ₹1,000, ₹1,000, ₹1,000, ₹1,500, ₹1,500, ₹1,500, ₹1,500, ₹2,000, ₹2,000, ₹2,500, ₹2,500, ₹2,500, ₹2,500, ₹3,000, ₹3,500, ₹4,000, ₹4,000

6. Binary Search

Binary search needs sorted data, so prices are first sorted with merge sort. The example below searches for a price of ₹2,500 (the app default).

Step	Low	High	Mid	Value at mid	Decision
1	0	31	15	₹1,000	search right half
2	16	31	23	₹2,000	search right half
3	24	31	27	₹2,500	match – found

Result: target found at index 27 after 3 comparison(s). With 1,000 items, about 10 steps would be enough.

7. Complexity summary

Algorithm	Time	Space	How it works
0/1 Knapsack	$O(nW)$	$O(nW)$	Fills a table of products × budget slots; each cell picks the better of include or exclude.
Binary search	$O(\log n)$	$O(1)$	Halves the sorted range on every comparison.
Merge sort	$O(n \log n)$	$O(n)$	Splits the list, sorts each half, then merges them in order.
Bubble / selection / insertion	$O(n^2)$	$O(1)$	Simple to follow but slow on large lists.

8. Notes

- Run-times in the Analytics page are measured live in the browser, so they vary by device and are not included here.
- Medicines in the catalogue are demo data; consult a pharmacist before taking any medicine.
- This PDF reproduces the app's catalogue, offers and algorithm outputs at the default settings; the interactive visualisations are not part of the PDF.